

Rakshya U. Sharma

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Professional Summary

Data analyst and bioinformatician with experience translating complex genomic, clinical, public-health, and research datasets into reproducible analyses and interpretable findings. Proficient in Python, R, statistical modeling, data visualization, workflow automation, and high-performance computing. Experienced collaborating with researchers, clinicians, and interdisciplinary teams to define analytical questions, evaluate data quality, communicate results, and support evidence-based decision-making.

Technical Skills

Programming and analysis: Python, R, Bash, SQL, pandas, NumPy, SciPy, scikit-learn, tidyverse, data.table

Statistics: Linear and logistic regression, association testing, multiple-testing correction, predictive modeling, dimensionality reduction, sensitivity analysis

Visualization and reporting: ggplot2, Matplotlib, R Markdown, Jupyter Notebook, scientific figures, technical reports, presentations

Data systems and reproducibility: Snakemake, Nextflow, Git/GitHub, Linux, Docker, Conda, YAML, HPC/SGE

Domain experience: Public health, health disparities, aging, genetics, clinical and phenotype data, biomedical research

Professional Experience

Bioinformatician

University of California, San Francisco

June 2024 – Present

San Francisco, CA

- Analyze large genetic, clinical, phenotype, and biomarker datasets to study modifiable risk factors and health outcomes in Alzheimer's disease and aging populations.
- Clean, harmonize, and validate data from more than 30 heterogeneous datasets, resolving inconsistencies in variable definitions, coding, sample identifiers, and analytical metadata.
- Apply regression modeling, association testing, polygenic risk-score analysis, multiple-testing correction, and sensitivity analyses to quantify relationships between health-related exposures and disease outcomes.
- Develop reproducible analytical workflows in Python, R, Snakemake, and Nextflow for data quality control, statistical analysis, and automated reporting in Linux and high-performance computing environments.
- Create figures, summary tables, and research presentations that translate technical findings for collaborators in genetics, epidemiology, neurology, and clinical research.
- Coordinate analyses across multiple research projects, document methods, track deliverables, troubleshoot technical issues, and communicate progress during interdisciplinary meetings.

Undergraduate Research Assistant

UC Santa Cruz Genomics Institute, Brooks Laboratory

April 2022 – June 2024

Santa Cruz, CA

- Developed computational methods to identify RNA-splicing patterns in large cancer transcriptomic datasets and connect molecular signatures with patient-level genomic and clinical variables.
- Performed data processing, statistical modeling, multiple-testing correction, validation, and visualization using Python, R, and reproducible workflows.
- Presented results through research reports, conference posters, and talks, translating complex analytical findings for scientific audiences.

Computational Genomics Intern

Broad Institute of MIT and Harvard

May 2022 – August 2022

Cambridge, MA

- Developed a computational pipeline to identify spinal muscular atrophy-related deletions and variants from large sequencing datasets.
- Evaluated analytical outputs, investigated discrepancies, visualized genomic evidence, documented methods, and contributed to a peer-reviewed publication.

Data Science Research Intern

Lawrence Berkeley National Laboratory

June 2021 – August 2021

Berkeley, CA

- Applied machine-learning and image-analysis methods to automate segmentation and quantitative feature extraction from plant-root images.
- Produced reproducible analytical outputs and presented findings at the Lawrence Berkeley National Laboratory research symposium.

Education

University of California, Santa Cruz

B.S. in Computer Science; Minor in Bioinformatics

June 2024

Santa Cruz, CA

Additional training: UCSF PROPEL Post-Baccalaureate Scholar; training in biomedical data science, research communication, scientific literature analysis, and interdisciplinary collaboration.